



Ahsanullah University of Science and Technology Bangladesh

COURSE OUTLINE

Part A

1. Course No./Course Code: **CSE CSE 4204 (BNQF 061)**
2. Course Title: **Computer Graphics lab**
3. Course Type (GE/Core Course/Elective): **Core Course**
4. Year/Level/Semester/Term: **Level 4 Term 2**
5. Academic Session: **Spring 2021**
6. Course Teacher/Instructor: **Mohammad Imrul Jubair, Md. Reasad Zaman Chowdhury**
7. Pre-requisite(s) (if any): **n/a**
8. Credit Value: **0.75**
9. Credit Hours: **1.5**
10. Total Marks: **100**
11. Rationale of the Course: **SDG 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all and SDG 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all**
12. Faculty: **Engineering**
13. Department: **Computer Science and Engineering (CSE)**
14. Programme: **Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE)**

15: Course Objectives:

- a. To learn graphics APIs, such as OpenGL/ WebGL; to write a simple program to basic primitives with coordinates and colors; to understand the basic concepts of shading language and its important parts: vertex and fragment shader; to understand the basic graphics pipeline.
- b. To learn basics about GLS, such as – different types of data used in GLSL such as attribute, uniform and varying; implementing mouse interaction in WebGL.
- c. To apply index buffer and to understand its necessity; to work with 3D objects and to use 3D transformation matrices
- d. To be able to apply the theoretical knowledge of viewing and projection in WebGL.
- e. TO apply the theoretical knowledge of texture in WebGL and implement basic animation in WebGL.
- f. To implement the theoretical knowledge of lighting and surface shading in WebGL and implement the Phong shading using GLSL.

16. Mapping of Course Outcomes with Bloom's Taxonomy and Programme Outcomes

Sl. No.	COs	POs	Bloom's Taxonomy		
			C	A	P
1	Comprehend the basics of computer graphics and the graphics pipeline	1			2
2	Apply 2D and 3D linear transformation to perform geometric transformation and viewing, and different rasterization techniques, shading models and textures.	2			3
3	Analyze different various algorithms used in computer graphics.	4			4

17. Mapping of COs with Knowledge Profiles, Complex Engineering Problem Solving and Complex Engineering Activities

Course Outcome	Knowledge Profile	Complex Problem Solving	Complex Engineering Activities
CO1	K3		

CO2	K6		
CO3	K4		

Part B

18. Week-wise Course Plan

Week	Topic	Teaching-Learning Strategy	Assessment Strategy	Corresponding COs
1	- Introduction to OpenGL/WebGL: Acquainting with the basics of OpenGL/WebGL, familiarizing with GLSL, Graphics pipeline and Shaders (vertex shader and fragment shader) - Drawing a point on the screen: Drawing a point	- <i>Lecture</i> - <i>Code Demonstration</i> - <i>Brain Storming Session</i>	<i>Assignments and online tests</i>	1
2	- Learning other drawing modes: Triangles, - Understanding different qualifiers in GLSL: Attributes, Uniforms, and Varying.	- <i>Lecture</i> - <i>Code Demonstration</i> - <i>Brain Storming Session</i>	<i>Assignments and online tests</i>	1,2
3	- Interactive Computer Graphics: Mouse and key input. - Transformations: Applying translation, rotation, and scaling using GLSL.	- <i>Lecture</i> - <i>Code Demonstration</i> - <i>Brain Storming Session</i>	<i>Midterm exam</i>	1,2
4	Model, View & Projection: Implementing camera and perspective projection.	- <i>Lecture</i> - <i>Code Demonstration</i> - <i>Brain Storming Session</i>	<i>Assignments and online tests</i>	2
5	Implementing texture mapping using GLSL.	- <i>Lecture</i> - <i>Code Demonstration</i> - <i>Brain Storming</i>	<i>Assignments and online tests</i>	2

		<i>Session</i>		
6	Lighting Model: Implementing Light models using GLSL.	- Lecture - Code Demonstration - Brain Storming Session	<i>Assignments and online tests</i>	2,3
7	Review class and project submission	-	<i>Lab Final Exam</i>	1,2,3

Part C

19. Assessment and Evaluation

1) Assessment Strategy: ***Class Performance, Viva, Assignments, Mid-Term Exam, Final Exam***

2) Marks distribution:

a) Continuous Assessment: ***Class Performance (20), Viva and Assignments (25), Online (10)***

b) Summative: ***Mid-term Exam (15), Final Exam (30)***

Part D

20. Learning Materials

20.1. Required (if any)

1. Fundamentals of Computer Graphics”, 4th Edition.
 Authored by: P. Shirley, S. Marschner
 Publisher: CRC Press
2. Computer Graphics Principles and Practice”, Second Edition in C, 2003.
 Authored by: J. D. Foley, A. Van Dam, S. K. Feiner and J. F. Hughes. Publisher: Pearson Education.

20.2. Recommended (if any)

20.3. Others (if any)

Prepared by:	Checked by:	Approved by:
Signature: <i>Reasad Zaman</i>	Signature:	Signature:
_____ Name: Md. Reasad Zaman Chowdhury Department: CSE Date: 24.3.2022	_____ Name: H M Zabir Haque OBE Program Coordinator, CSE Date:	_____ Name: Dr. Mohammad Shafiul Alam HOD, CSE Date:

Annex-1: PEO of CSE

PEO1 - Professionalism

Graduates will demonstrate sound professionalism in computer science and engineering or related fields.

PEO2 – Continuous Personal Development

Graduates will engage in life-long learning in multi-disciplinary fields for industrial and academic careers.

PEO3 – Sustainable Development

Graduates will promote sustainable development at local and international levels.

Annex-2: Mapping of PEO-PO

	PEO1	PEO2	PEO3
PO1 - Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.	√		
PO2 - Problem analysis: Identify, formulate, research and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.	√		
PO3 - Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.	√		
PO4 – Investigation: Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.	√		
PO5 - Modern tool usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.	√		
PO6 - The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.	√		√

PO7 - Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.	√		√
PO8 – Ethics: Apply ethical principles and commit to professional ethics, responsibilities and the norms of engineering practice.	√		
PO9 - Individual work and teamwork: Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings.	√	√	
PO10 – Communication: Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.	√		
PO11 - Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work as a member or a leader of a team to manage projects in multidisciplinary environments.	√		
PO12 - Life-long learning: Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.		√	

Annex-3: Blooms Taxonomy *

Level	Cognitive Domain – Revised Version	Affective Domain	Psychomotor Domain
1	Remember (1)	Receiving Phenomena (1)	Perception (1)
2	Comprehend (2)	Responding to Phenomena (2)	Set (2)
3	Apply (3)	Valuing (3)	Guided Response (3)
4	Analyse (4)	Organizing Values (4)	Mechanism (4)
5	Evaluate (5)	Internalising Values (5)	Complex Overt Response (5)
6	Create (6)		Adaption (6)
			Origination (7)

* Based on “REVISED BLOOM’S TAXONOMY INDICATOR v3.31” , available at <http://adept.mmu.edu.my/wp-content/uploads/2018/09/Blooms-Taxonomy-Indicator-v3.31.xls>